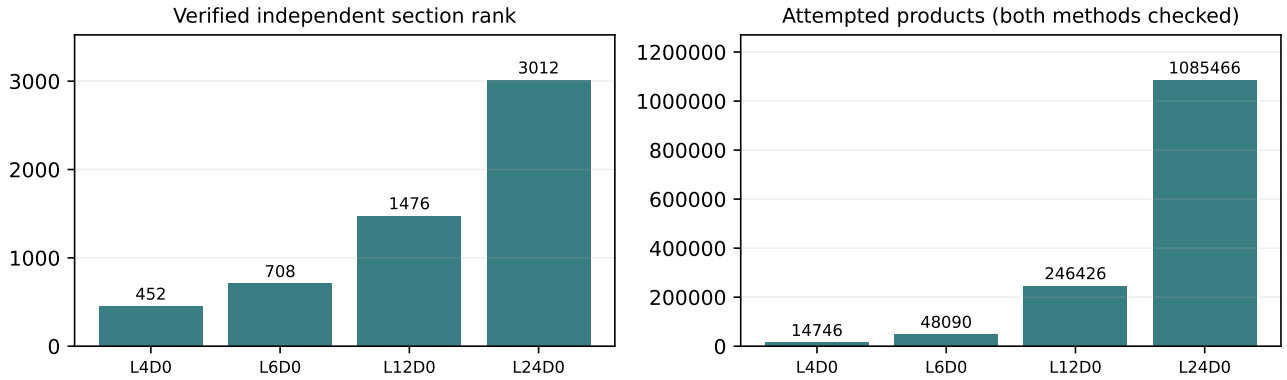
R43: actual larger source and section spaces

Observations report: three accepted native computations, one accepted Lean workflow and two retained rejected attempts. No independent evidence verdict or index-calculus performance comparison.



multiply43-01; rank stopping rule fixed before data. Source expansions: jets43-02.

Source-bound precision615 expansion

At the certified degree5point Q, construct all615coefficients of u,v,b,c,lambda,nu. Every Newton stage checks the five actual source equations and preserves old coefficients. All eight units/inverses, five coefficient-descent conjugates and closure pass; a perturbed top nu coefficient is rejected.

All610saved reference rows agree through precision154: six coordinates, eight units, eight inverses,68sections,196quadratics,324cubics. Grouped and direct monomial evaluation agree for204original/twist values through154. The actual descended functions agree with both original orientations at615. The old section ranks68/196/324are recovered in dimension3075.

Larger complete product bases

Space	Input rank	Output rank	Products independently checked
L4D0	196	452	14746
L6D0	324	708	48090
L12D0	708	1476	246426
L24D0	1476	3012	1085466

All attempted products and pivots reconstruct from selected original columns. Each selected function has its parent pair and full615coefficients. Enumeration stops upon reaching the prescribed dimension; it is not an all-pair census.

Completeness bridge, proof scope and remaining work

The retained proper source has genus61and D0has degree128. Products satisfy the indicated pole bounds. Evaluation at615Qhas degree3075, exceeding the largest pole bound3072. Thus evaluation is injective under the retained proper/pole bridge. Each independent native rank meets its Riemann–Roch upper bound. This yields complete spaces under that bridge, without assuming surjectivity of an untested product pool.

With D1=4D0these are the requested L(D1),L(3D1),L(6D1)dimensions452,1476,3012. This removes a concrete setup mismatch for ordinary367/460divisors; it does not construct or evaluate those complete fibers.

Lean and retained errors

Three named mathematical theorems prove characteristic-two AS telescoping, the ten-term solution under nilpotence615, and coefficient extraction at615. The last proof is reused from R38. Audit also includes three generated recursion equations: six declarations total, zero failing/trusted. No native evaluation. Native arrays, matrix ranks, proper normalization and RR completeness are not formalized by these generic lemmas.

jets43-01failed in Rust compilation because a series edit targeted Field instead of Jet; no native trial ran. lean43-01failed an implicit characteristic two cast tactic. Committed repairs used fresh attempts with unchanged caps. No accepted producer reran. Native build/run caps:4GiBvirtual/1200seconds; Lean4.34.1/mathlibd13f23b,one job,4096MiBheap,1200seconds compile/audit.

Next actual constructions

Complete ordinary367/460fibers and exceptional multiplicities; certify padding disjointness; construct complete small and large vanishing spaces inside the new bases. Then extend full-generator division/class evaluation and check actual factor images. Never invert a jet vanishing on padding.

Proper level77target binding, the chosen Koblitz-model isogeny, full quotient saturation/polarization, node187 coupled valuations and theta descent remain separate. There is no evaluated larger-n transfer, IC relation rank, unknown-scalar recovery or speedup. Independent scientific/semantic review remains pending; artifact byte verification is a separate check.

Sources: retained R27/R34/R35/R38/R40 and Khuri-Makdisi math/0409209v3 representation-B setup, plus Stacks01V4; links in LITERATURE.md. Full raw formats, matrices, attempted product coordinates and lineage are documented in FORMAT.md and the content-addressed artifact.